

REVIEW

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# Role of big data analysis to improve the efficiency of healthcare system using machine learning techniques

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## Abstract

Big data analytics is reshaping healthcare by integrating machine learning with large medical datasets. It enables predictive diagnosis, personalized treatment, and real-time monitoring through telemedicine and wearable devices, improving decision-making and extending care to underserved populations. This survey reviews applications of big data in enhancing diagnosis accuracy, chronic disease management, and resource optimization. Predictive models for heart disease and stroke demonstrate significant improvements in timely intervention and patient outcomes. However, challenges such as poor data quality, privacy concerns, interoperability issues, and connectivity gaps hinder widespread adoption. Ethical considerations around patient confidentiality also remain critical. The paper compares existing big data technologies and machine learning frameworks, highlighting their strengths, limitations, and clinical impact. It concludes with recommendations for advancing predictive analytics, personalized medicine, and scalable, data-driven healthcare systems to guide researchers, practitioners, and policymakers.

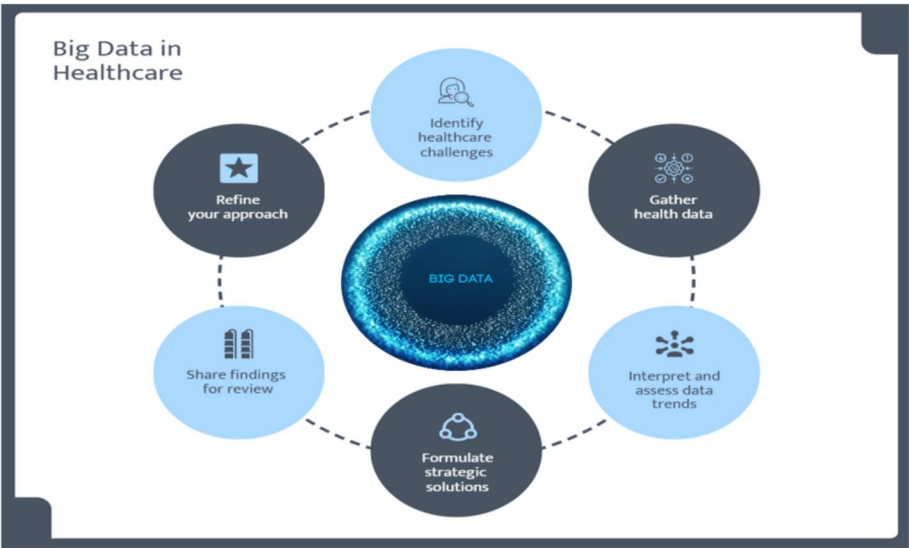
**Keywords** Big data analytics, Machine learning, Supervised learning, Healthcare system, Disease prediction

## 1 Introduction

The current healthcare industry has seen a dramatic transition as a result of the integration of cutting-edge technology, with big data analytics emerging as a game-changer. The huge amount of data that is generated by healthcare organizations or system offers a particular perspective to improve the patient's hospitality, reschedule trials, advanced medical examination, etc. To generate significant data insights, patterns, and associations from huge and varied datasets a big. data analytics in healthcare system needs a back breaking analysis of the data. The assimilation of big data analysis into the healthcare system increases the patient consequences, make best use of resources, improve decision-making, etc. [1]. An overview of the applications of big data in the healthcare industry is illustrated in Fig. 1.



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**Fig. 1** Big data analytics in healthcare (created by the author)

**Table 1** Healthcare data sources and associated bottlenecks

| S. No. | Data sources                      | Challenges/bottlenecks                                                             | Refs. |
|--------|-----------------------------------|------------------------------------------------------------------------------------|-------|
| 1      | Electronic health records (EHRs)  | Large volume, inconsistent quality, and interoperability issues                    | [3]   |
| 2      | Wearable devices & IoT sensors    | High-velocity data streams, noise, and poor data quality                           | [4]   |
| 3      | Medical imaging                   | Massive file sizes, storage burden, and complex feature extraction                 | [5]   |
| 4      | Genomic databases                 | High dimensionality, computational complexity, and privacy concerns                | [6]   |
| 5      | Real-time ICU/clinical monitoring | Continuous data flow, latency in integration, difficulty in timely decision-making | [7]   |
| 6      | Patient histories & demographics  | Unstructured formats, missing values, and lack of standardization                  | [7]   |

- (i) **The Healthcare Data Flooding:** The exponential growth of healthcare data from diverse sources such as electronic health records (EHRs), wearable sensors, imaging technologies, genomic databases, and real-time clinical monitoring has resulted in a significant data flood. While these datasets provide valuable opportunities for improving patient care, their massive volume and heterogeneity exceed the processing capabilities of traditional analytical methods [2].
- Key challenges emerging from this data landscape include difficulties in handling large-scale, diverse datasets, inefficiencies in extracting actionable insights, limited predictive modeling capacity for early diagnosis and treatment outcomes, and obstacles in optimizing healthcare resource allocation. These limitations underscore the necessity for advanced big data analytics and machine learning techniques to transform raw data into meaningful, predictive, and operational insights for healthcare. As summarized in Table 1, diverse data sources contribute to the phenomenon of healthcare data flooding, while bottlenecks such as heterogeneity, weak predictive capacity, and inefficient resource allocation highlight the urgent need for advanced analytics-driven solutions.
- (ii) **Enhancing Clinical Decision:** Big data analytics play an important role to empower the healthcare personnel which deals rapid and particular insights to enhance

medical decision making. With the help of patient credentials, historical data, clinical treatment consequences, graphical statistics, medical professionals may take better clinical decisions. This approach changes a great precision of analysis and patient medical diagnosis during dealing a particular treatment [8].

- (iii) **Personalized Medicine and Precision Healthcare:** The idea of customized treatment is among the biggest advances in big data analytics in the field of healthcare. Healthcare professionals can find patterns and connections that guide individualized treatment plans by examining big datasets that contain genetic data, lifestyle variables, and therapy responses. Precision medicine has the potential to maximize positive treatment results while reducing negative ones. The healthcare industry is responsible for generating a humongous volume of heterogeneous data from various sources like electronic health records (EHRs), medical imaging equipment, wearable sensors tracking health metrics, and large genomic databases. As much as rich data is made available to the industry, effective utilization of the accumulated data remains a continuous, serious challenge due to several major factors [9]:
  - (a) **Big Data Analytics** can effectively transform raw healthcare data into actionable insights that can feed into enhanced decision-making processes within the industry [10].
  - (b) **Machine Learning Techniques** can significantly enhance predictive modeling capabilities, thereby allowing early disease detection and the formulation of personalized treatment plans tailored to individual patient needs [11].
  - (c) **Data-Driven Techniques** can be utilized to optimize operational efficiency, improve overall patient outcomes, and enable a more proactive approach to the management practices of healthcare [11].

The overall aim of this survey is to provide a comprehensive and extensive review that goes in-depth into the integration of machine learning strategies and big data analytics within the healthcare industry. The specific aims of this research study are outlined and listed as follows:

- (a) **To Review the Role of Big Data in Healthcare:** This involves a comprehensive review and analysis of the pivotal role that big data plays in the improvement of patient care, improving clinical decision-making processes, and optimizing operational efficiency in different healthcare systems and environments.
- (b) **To Recognize Challenges in Utilization of Healthcare Data:** This aim involves an analysis of the intricate challenges and hurdles that are inherently associated with issues related to data volumes, variability in the quality of data, security threats, and issues of integration, which all contribute to hindering the effective deployment of big data strategies within the healthcare environment.
- (c) **To Examine Predictive Frameworks Based on Machine Learning:** Here, we will perform a review of the prevailing predictive frameworks that are based on machine learning technologies, with special focus on their application for the early detection of diseases and estimation of related risks through the application of modern-day healthcare datasets.
- (d) **To Examine the Performance of Machine Learning Algorithms:** This aim involves a comparison and contrast of the performance measures of a range of different machine

learning algorithms, particularly in the context of their accuracy, precision, and reliability in the context of healthcare data analysis and evaluation.

- (e) To Provide Inputs for Future Innovations in Healthcare: This involves providing valuable insights and suggestions on how advanced data analytics strategies can be effectively integrated within healthcare systems, which, in turn, will lead to innovations in fields such as personalized medicine, telemedicine, and proactive patient care programs.

The adoption of big data analytics (BDA) in healthcare has turned into a global phenomenon, revolutionizing medical practice, policy-making, and patient care. In the United States, big data and machine learning solutions in clinical workflows have enabled predictive modeling of early disease detection, thereby reducing the hospital readmission rate by up to 20% [12]. In the European continent, countries such as the United Kingdom and Germany have utilized national health databases to maximize resource planning, improve management of chronic diseases, and extend precision medicine programs [13]. On the Asian–Pacific continent, Chinese, Indian, and Singaporean mega-projects have leveraged big data platforms to analyze genomic information, track outbreaks of infectious diseases, and improve telemedicine services in rural settings [14]. According to a report by the World Health Organization (WHO), healthcare systems across the globe are experiencing unprecedented acceleration of heterogeneous clinical data driven by the revolution in medical imaging, electronic health records (EHRs), wearable sensors, and genomics research [15]. This explosive growth underlines the compelling imperative for more advanced analytical methods, such as machine learning, to convert vast and heterogeneous datasets into actionable information that enhances the efficiency and quality of healthcare across the world [16].

The healthcare sector was selected as the topic of this research due to its critical function in society, its contribution to the economy, and its growing reliance on evidence-based decision-making. Healthcare expenditure globally accounts for approximately 10% of global GDP, and the expenditure is anticipated to increase exponentially due to aging populations, the prevalence of chronic disease, and the demand for personalized therapies [17]. The COVID-19 pandemic has also highlighted the need to provide rapid, data-driven responses to public health crises, emphasizing the need to combine big data analytics and machine learning to facilitate disease surveillance, outbreak forecasting, and resource planning [18]. The sector also generates one of the largest and most heterogeneous sources of data—ranging from electronic health records and genomic data to medical imaging and real-time patient monitoring—that present unique opportunities and challenges for analysis [19]. All these factors combined make healthcare an industry of high priority for investigating the potential of big data analytics and machine learning to improve operational efficiency, clinical outcomes, and public health in general.

Recent research has laid the groundwork for contemporary research, which has contributed towards the application of machine learning and big data analytics in the healthcare sector. Mavrogiorgou et al. [20] introduced an exhaustive list of machine learning algorithms applicable in healthcare risk prediction, highlighting the potential of algorithmic techniques towards improved early diagnosis and patient care. Kiourtis et al. [21] suggested a cross-sector data space strategy combining human health and environmental risk factors, highlighting the broader potential of big data systems to integrate heterogeneous data sets to achieve improved public health returns. Zafeiropoulos

et al. [22] explored interpretable stroke risk prediction models by leveraging machine learning, highlighting transparency and explainability towards clinical decision support systems. These contributions collectively highlight the value of combining large-scale data processing with higher-level analytics to address predictive accuracy as well as the operational requirements of modern healthcare systems.

This research makes several significant contributions to the existing literature. First, it makes a thorough synthesis of how big data analytics, when applied with machine learning techniques, can enhance predictive modeling, improve clinical decision-making, and maximize resource utilization in healthcare systems. Second, it showcases cross-fertilization of ideas from global application as well as cutting-edge state-of-the-art research, thus filling the gap between theoretical development and practical application [23]. Third, the research highlights domain-specific applications such as personalized medicine, early stage detection of diseases, and enhancement of operational efficiency while highlighting the significance of interpretability and transparency of predictive models [24]. However, this research has its limitations. Focus is restricted to the application of machine learning-based big data analytics and does not incorporate other evolving technologies such as blockchain or IoT-based security protocols. Second, though the review includes a thorough body of literature, it quotes mainly English-language and peer-reviewed literature and thus potentially excludes regional literature studies as well as non-indexed contributions. These significant points and limitation provide both a specific research question and the scope for further research to take the findings to other related interdisciplinary areas.

The remainder of this paper adopts the following organization. Section II presents an in-depth background of the fundamental principles of big data analytics and machine learning algorithms best suited to healthcare use cases. Section III presents recent advancements and milestone case studies illustrating the convergence of these technologies for healthcare use cases. Section IV presents the primary challenges, including heterogeneity, scalability, and ethics, and potential solutions presented in the literature. Section V contrasts the performance of some machine learning algorithms when applied to large-scale healthcare datasets. Section VI presents future research directions, signaling the contribution of new technologies and interdisciplinary research for advancing healthcare efficiency. Finally, Section VII concludes the paper with a summary of findings and practical implications for stakeholders in the healthcare ecosystem [25].

### 1.1 Research gap and novelty

Although several reviews have explored the role of big data analytics and machine learning in healthcare, most of them are limited in scope. Some focus mainly on technical algorithms without connecting them to clinical practices, while others emphasize healthcare applications but lack a comparative analysis of machine learning frameworks. In addition, existing surveys often address either predictive analytics or personalized medicine in isolation, leaving a gap in understanding their combined role in improving healthcare efficiency at both clinical and operational levels.

This survey contributes a novel synthesis by bridging these gaps. Unlike prior reviews, it integrates perspectives from global healthcare applications, technical frameworks, and domain-specific use cases such as telemedicine, real-time monitoring, and precision medicine. It also provides a structured comparison of big data tools and machine

learning models, emphasizing their strengths, limitations, and clinical relevance. Furthermore, the paper highlights interpretability, scalability, and ethical considerations as essential dimensions often overlooked in earlier works. By doing so, this study offers researchers, practitioners, and policymakers a comprehensive roadmap for adopting data-driven healthcare solutions.

## 2 Literature review

Understanding a firm or organization's degree of productivity, competitiveness, and innovation requires an understanding of big data. Big data analytics has been investigated by several industries, including telecommunications, manufacturing, retail, healthcare, and the public sector [26]. Big data integration, digitalization, and use may benefit a wide range of businesses, including multi-providers and large healthcare providers, as well as tiny startups and global firms. With the increasing usage of electronic health records (EHRs) and patient monitoring systems, enormous amounts of data, including physiological data, have been continuously flowing and accumulating, requiring mining and analysis [27]. The authors also point out that in order to give information on uncommon diseases that would have been difficult to evaluate without high sample sizes, there is interest in compiling and integrating EHR and other patient-related data in national databases. This is because big data's promise in healthcare is becoming more widely acknowledged [28]. Furthermore, keep in mind that when health providers digitized their services, they gathered a lot of data in order to either better serve their patients or adhere to legal requirements. Research has also shown that the amount of data and information in the area is anticipated to increase over time as technology is employed to enable healthcare performances through the usage of significant and essential information within the healthcare sectors [29]. The sources of the patient data are listed and grouped, and this data is being collected from a variety of sources.

### 2.1 Overview of big data

Big Data is a phrase that refers to datasets that not only occur in huge numbers but are also immensely complex in nature such that conventional data processing mechanisms are incapable of handling them in an efficient and effective manner. The importance of Big Data has increased exponentially year after year during the past couple of years, especially in industries like healthcare [30], where an unending stream of data is generated from multiple sources like electronic health records, medical imaging devices, wearable health monitoring systems, and genomics research.

### 2.2 Evolution of big data

Big Data first came into existence in the early 1990s, slowly gaining pace and momentum through the work of industry experts like John R. Mashey. The initial focus of data handling was structured information that occurred within relational database management systems (RDBMS). This involved employing methods like Structured Query Language (SQL) and Extraction, Transformation, and Loading (ETL) processes to manipulate and handle data. But with technological advancements, the domain of Big Data grew phenomenally large to encompass unstructured content on the web, which drew individuals to create new storage and processing mechanisms to address the complexity specifically. In addition to this, the extensive usage of mobile devices and numerous sensors has led



to the huge explosion in data generation, which further gave rise to the development of new methods designed to handle the increased volume, variety, and velocity of data efficiently [11].

### 2.3 Types of big data analysis

Big Data analysis is classified into four broad categories, with each type of analysis having its own specific purpose and objectives and these types are illustrated in Fig. 2.

- (a) Descriptive Analytics: This specific form of analysis is mainly concerned with extracting the summary of historical data to understand and grasp what has happened during a time span. It utilizes various techniques like data aggregation and data mining to infer meaningful information about past occurrences and trends.
- (b) Diagnostic Analytics: After gaining the initial insights through descriptive analytics, diagnostic analytics seeks to know the reason for the occurrence of certain events. This type of analysis delves deeper into the data to uncover underlying patterns and relationships and tends to take advantage of advanced techniques like drill-down and data discovery to achieve deeper insights [31].
- (c) Predictive Analytics: This type utilizes a blend of statistical models and machine learning algorithms for the sake of forecasting potential future events, based on historical data as its foundation. The sole purpose here is to forecast upcoming trends and behavior and hence equip organizations with meaningful insights that allow proactive decision-making in expectation of the developments [31].
- (d) Prescriptive Analytics: The most advanced and mature type of analytics is referred to as prescriptive analytics. This amazing technique not only forecasts what future results can be expected to happen but also actively recommends specific steps that can be taken in order to bring about the desired outcome. By combining predictive models with advanced optimization algorithms, it can recommend specific strategies that can be used by organizations to navigate their way to success.

### 2.4 The four V's of big data

Big Data is generally described by four basic dimensions that are usually referred to as the Four V's [32]:



**Fig. 2** Types of big data analysis (created by author)

- (a) Volume: This is the term used to describe the phenomenally massive amounts of data that are produced every single second, which demand scalable storage systems that can handle such humongous amounts of data.
- (b) Velocity: This dimension refers to the extremely fast rate at which new data is created, as well as the rate at which this data needs to be processed in order to be considered useful and actionable.
- (c) Variety: This dimension refers to the various forms of data that exist, ranging from structured data to unstructured data to semi-structured data, all of which come from a humongous number of different sources.
- (d) Veracity: This dimension refers to the quality and reliability of the data itself, addressing the inherent issues related to data uncertainty and the genuineness of the information being processed.

## 2.5 Tools used in big data analytics

In Big Data analytics, numerous tools have been developed to facilitate and ease the process of data analysis of big data. Each tool has been developed to address and manage specific data processing aspects and the resulting analysis that follows. To provide a summary and insight, the Table 2 provides a list of some of the most visible tools utilized in the market at present, along with a brief description of their main functions and features:

## 2.6 Related work

Recent advances in big data analytics (BDA) with machine learning (ML) have introduced new innovations in healthcare applications from predictive modeling to personalized treatment recommendations. Various studies have illustrated the application of BDA in healthcare using different data sources, methodology, and performance metrics. For instance, some studies illustrate risk prediction using structured clinical data, while others add environmental or lifestyle data for improved model performance. Model explainability has also been another area of clinical utility with some studies pointing towards explainable AI architectures for transparency in decision-making. In spite of such studies, variations continue to remain in dataset diversity, scalability, algorithmic complexity, and performance metrics. To facilitate easy systematic comparison of such studies, Table 3 presents an overview of the relevant characteristics of representative studies, e.g., application areas, data sources, ML algorithms, performance measures, limitations, and number of citations for easy reference. Not just does the comparison illustrate the methodological diversity of the literature, but also the research gap being filled by the current research.

**Table 2** Tools used in big data analytics

| Tool         | Objective                             | Key characteristics                        |
|--------------|---------------------------------------|--------------------------------------------|
| Hadoop       | Decentralized data processing         | Expandability, fault tolerance             |
| Apache Spark | Instantaneous data analysis           | In-memory processing, swift computation    |
| Tableau      | Data representation                   | Interactive dashboards, smooth integration |
| RapidMiner   | Machine learning and data exploration | Drag-and-drop interface, model deployment  |
| KNIME        | Data unification and examination      | Workflow-based analysis, open-source       |
| MongoDB      | NoSQL data management                 | Adaptable schema, high scalability         |
| Cassandra    | Decentralized database management     | High availability, linear scalability      |
| Power BI     | Business intelligence and reporting   | Data visualization, AI integration         |



**Table 3** Summary of related research works in big data analytics for healthcare

| References | Study focus/ap-<br>plication domain                          | Dataset type and<br>source                                     | Machine learn-<br>ing algorithms                               | Perfor-<br>mance<br>metrics             | Reported<br>limitations                                             |
|------------|--------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------|
| [20]       | Risk prediction in<br>healthcare                             | Electronic health<br>records (EHRs) from<br>multiple hospitals | Logistic regres-<br>sion, random<br>forest, neural<br>networks | Accuracy,<br>precision,<br>recall       | Limited gen-<br>eralizability to<br>non-hospital<br>datasets        |
| [21]       | Environmental risk<br>correlation with<br>health data        | Environmen-<br>tal monitoring<br>data + patient<br>records     | Decision trees,<br>gradient boosting                           | F1-score,<br>ROC-AUC                    | Integration com-<br>plexity due to<br>heterogeneous<br>data sources |
| [22]       | Stroke risk predic-<br>tion with interpre-<br>table models   | Stroke patient<br>dataset with clinical<br>attributes          | XGBoost with<br>SHAP-based<br>interpretability                 | ROC-AUC,<br>Sensitivity,<br>Specificity | Limited sample<br>size and geo-<br>graphical scope                  |
| [23]       | Predictive model-<br>ing in precision<br>medicine            | Genomic data + clin-<br>ical trial records                     | Support vector<br>machines, deep<br>neural networks            | Accuracy,<br>ROC-AUC                    | High com-<br>putational<br>requirements                             |
| [24]       | General ML ap-<br>plications in clinical<br>decision support | Mixed datasets<br>(EHR, imaging, wear-<br>able devices)        | CNNs, RNNs, en-<br>semble methods                              | Accuracy,<br>F1-score                   | Black-box nature<br>affecting model<br>transparency                 |

## 2.7 Previous work

Big data analytics and machine learning have seen a lot of research in healthcare. Many studies only work on disease prediction by using algorithms such as SVM, RF, DT, and neural network, with high accurate rates in discovering heart disease, diabetes and so on; as well as cancer. The researchers are focusing upon various factors of the classification ceiling, investigating the factors which will improve predictive performance, i.e., feature selection methods, and improvements in predictive performance that were capabilities of mobile health applications as well as data integration technologies. Furthermore, big data's rapid growth in health care has been examined with emphasis on its importance during a global health crisis such as COVID-19 and its applications in early detection of disease, personal medicine, and resource optimization. These studies together highlight the potential for big data and machine learning to facilitate improvements in patient care. Table 4 summarizes key contributions from previous research:

## 2.8 A comparative analysis of traditional versus AI and ML-based healthcare systems

Traditional healthcare systems rely primarily on human processes that rely on the competence and expertise of doctors, supplemented by rule-based diagnostics that inform medical decisions. Traditional systems such as these are typically limited by the frailties of human judgment and tend to embrace reactive approaches that are primarily focused on symptom treatment once it has occurred. Such a reliance on human decision-making can unfortunately lead to misdiagnosis, delayed treatments, and an inefficient allocation of available resources in the healthcare system. By contrast, AI and ML-based systems rely on large datasets sourced from electronic health records, advanced medical imaging technology, and new wearable devices. This convergence enables sophisticated predictive analytics, enables early disease identification, and facilitates the creation of personalized treatment plans based on individual patient requirements. AI and ML-based systems greatly improve the accuracy of diagnoses, improve the efficiency of operations through various forms of automation techniques, and enable a proactive model of healthcare. A comparative analysis is shown below with help of Table 5 Ultimately,

**Table 4** Previous works

| S. No. | Author(s)        | Year | ML method/approach                                 | Application area                             | Key findings                                                              | Refs. |
|--------|------------------|------|----------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------|-------|
| 1      | Evans et al      | 2015 | Conceptual framework (smart health)                | Integrated health-care platforms             | Proposed unified learning system for healthcare integration               | [33]  |
| 2      | Pan et al        | 2022 | Comparative study of ML algorithms on UCI datasets | Predictive modeling                          | Categorical features improved prediction over numerical features          | [34]  |
| 3      | Reddy et al      | 2024 | Gradient Boosting, Random Forest                   | Disease forecasting (diabetes, liver, heart) | Achieved effective multi-disease predictions using ensemble methods       | [35]  |
| 4      | Shvachko et al   | 2010 | Big Data methodologies                             | Digital health data analysis                 | Highlighted need for robust methods to handle large-scale healthcare data | [36]  |
| 5      | Rahul et al      | 2023 | Big Data analysis and visualization methods        | Outlier detection, industrial & healthcare   | Showcased big data's role in improving anomaly detection                  | [37]  |
| 6      | Guyon et al      | 2016 | Support Vector Machine (SVM)                       | Disease prediction                           | Achieved 90.6% accuracy with optimized parameters                         | [38]  |
| 7      | Bakkery et al    | 2020 | Deep Learning                                      | Breast cancer survival prediction            | Improved treatment decision-making through survival prediction            | [39]  |
| 8      | Okereke et al    | 2023 | Mobile application framework                       | COVID-19 & pandemic management               | Supported healthcare providers and governments globally                   | [40]  |
| 9      | Bataineh et al   | 2022 | Hybrid MLP + PSO model                             | Heart disease detection                      | Surpassed earlier models with 84.61% accuracy                             | [41]  |
| 10     | Sterling et al   | 2017 | Survey study                                       | Big data in healthcare                       | Highlighted transformative potential of BDA across domains                | [42]  |
| 11     | Calvert et al    | 2015 | ML + big data + blockchain                         | Public health surveillance                   | Demonstrated predictive capabilities for disease outbreaks                | [43]  |
| 12     | Srinivasan et al | 2023 | Neural networks                                    | Heart disease prediction (UCI datasets)      | Achieved 98.7% accuracy                                                   | [44]  |
| 13     | Rehman et al     | 2022 | ML on DNA microarray data                          | Cancer classification                        | Identified key genomic factors for early diagnosis                        | [45]  |
| 14     | Ozcan et al      | 2023 | CART algorithm                                     | Heart disease prediction                     | Reached 87% accuracy                                                      | [46]  |
| 15     | Yousef et al     | 2021 | Data integration strategies                        | Healthcare data management                   | Addressed integration and management of large healthcare datasets         | [47]  |
| 16     | Uddin et al      | 2020 | Review                                             | COVID-19 analytics                           | Emphasized essential role of BDA during pandemic                          | [48]  |
| 17     | Pathan et al     | 2022 | Feature selection (ANOVA-F test)                   | ML model optimization                        | Careful feature selection enhanced prediction accuracy                    | [49]  |
| 18     | Islam et al      | 2024 | Review of ML techniques (SVM, RF, DT, NB)          | Chronic disease prediction                   | Suggested hybrid approaches to improve predictive performance             | [50]  |
| 19     | Pelegri et al    | 2010 | Hadoop-based Big Data system                       | Early illness detection                      | Proposed scalable big data system for healthcare                          | [51]  |
| 20     | Zhang et al      | 2016 | Review                                             | Big data analytics in healthcare             | Provided advances and future research directions                          | [52]  |
| 21     | Greasley et al   | 2019 | Prediction models                                  | Brain disease detection                      | Detected diseases before symptom onset                                    | [53]  |
| 22     | Grover et al     | 2016 | Systematic review (205 studies)                    | BDA applications in healthcare               | Highlighted wide-ranging healthcare applications                          | [54]  |
| 23     | Bachiri et al    | 2016 | Big Data approaches                                | Pharmacy & toxicology                        | Applied BDA for critical challenges in toxicology and pharmacy            | [55]  |

**Table 4** (continued)

| S. No. | Author(s)     | Year | ML method/approach     | Application area       | Key findings                                                | Refs. |
|--------|---------------|------|------------------------|------------------------|-------------------------------------------------------------|-------|
| 24     | Benhlma et al | 2018 | Data mining techniques | Healthcare analytics   | Discussed evolution of data mining in healthcare            | [56]  |
| 25     | Wang et al    | 2015 | Predictive analytics   | Cardiovascular disease | Stressed real-time data's role in early infection detection | [57]  |

**Table 5** A comparative analysis of traditional versus AI and ML-based healthcare systems [58, 59]

| Aspect                    | Traditional healthcare system                           | AI & ML-based healthcare system                                                          |
|---------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------|
| Approach                  | Reactive, symptom-driven                                | Proactive, anticipatory, and data-informed                                               |
| Decision-making           | Dependent on physician expertise and clinical protocols | Enhanced by data analysis, forecasting models, and real-time data                        |
| Data utilization          | Confined to structured data and patient history         | Leverages vast amounts of structured and unstructured data from diverse sources          |
| Accuracy                  | Prone to human error and inconsistency                  | High diagnostic precision through pattern recognition and anomaly detection              |
| Personalization           | Standardized treatment plans based on general protocols | Tailored treatment based on individual data and predictive analytics                     |
| Efficiency                | Labor-intensive manual procedures                       | Streamlined workflows, faster diagnoses, and optimized resource allocation               |
| Scalability               | Constrained by human resources and infrastructure       | Exceptionally scalable with advanced computational capabilities                          |
| Resource management       | Manual distribution of medical resources                | Optimized via predictive analytics for demand forecasting                                |
| Real-time monitoring      | Limited to in-hospital tracking                         | Ongoing monitoring via IoT devices and wearable technology                               |
| Administrative workload   | Significant administrative load on healthcare providers | Decreased burden with automation of routine tasks                                        |
| Error rate                | Higher likelihood of diagnostic mistakes and oversights | Reduced error rate through data-driven support for decisions                             |
| Adaptability              | Gradual adaptation to evolving medical knowledge        | Rapid evolution through continuous learning and data analysis                            |
| Cost efficiency           | Elevated operational costs due to manual tasks          | Economically viable long-term through automation and predictive upkeep                   |
| Response to health crises | Reactive, with delayed outbreak detection               | Predictive modeling facilitates early detection and swift response to health emergencies |

these advancements result in better patient outcomes and a more efficient optimization of resources in the healthcare industry.

## 2.9 Role of machine learning in clinical decision-making

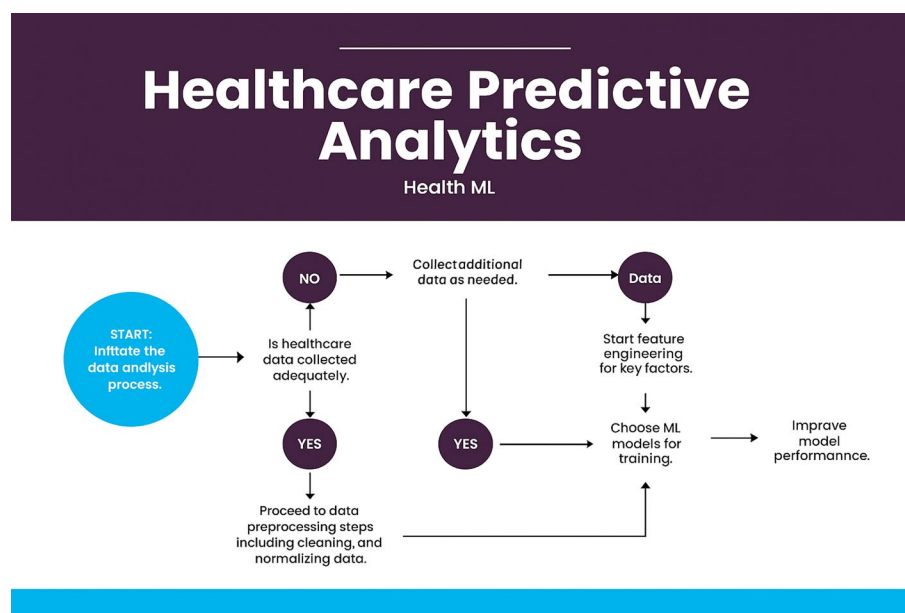
Machine learning (ML) is revolutionizing and redefining the face of clinical decision-making by enabling healthcare professionals to make highly accurate, data-driven decisions based on a vast treasure trove of data. By carefully analyzing huge amounts of data drawn from electronic health records (EHRs), medical imaging devices, genomic data, and real-time patient monitoring devices, ML algorithms can identify intricate patterns and make accurate predictions for a variety of health outcomes with high accuracy. This revolutionary ability not only enhances diagnostic accuracy but also enables the early identification of diseases, while aiding in performing comprehensive risk assessments for serious health conditions like heart disease, cancer, and diabetes [60]. Furthermore, ML models also offer customized treatment recommendations personalized uniquely for individual patients based on significant parameters like a patient's history, genetic profile, and reactions to previous treatments. In critical care environments, the

use of ML algorithms is imperative because they are responsible for predicting patient deterioration, optimizing resource allocation, and enabling timely medical intervention to be made available when required. Furthermore, clinical decision support systems (CDSS) based on machine learning play a significant role in preventing diagnostic errors, improving overall patient safety, and optimizing workflow efficiency by providing real-time insights augmented with evidence-based recommendations. With the domain of machine learning promising to expand further and develop itself, its integration into healthcare on a seamless scale promises much towards further streamlining clinical decision-making processes, enhancing patient outcomes significantly, and ultimately redefining the future of medical practice for the better.

### 3 Role of bigdata in predictive analytics for disease detection

By using predictive analytics algorithms, healthcare providers can identify populations at risk, anticipate disease outbreaks, and take action before conditions worsen. This proactive approach not only improves patient outcomes but also lowers the burden of preventable diseases, which increases overall system efficiency [61]. Big data analytics helps healthcare systems make this shift from reactive to proactive care models. Figure 3 presents the flow chart of big data working in predictive analysis.

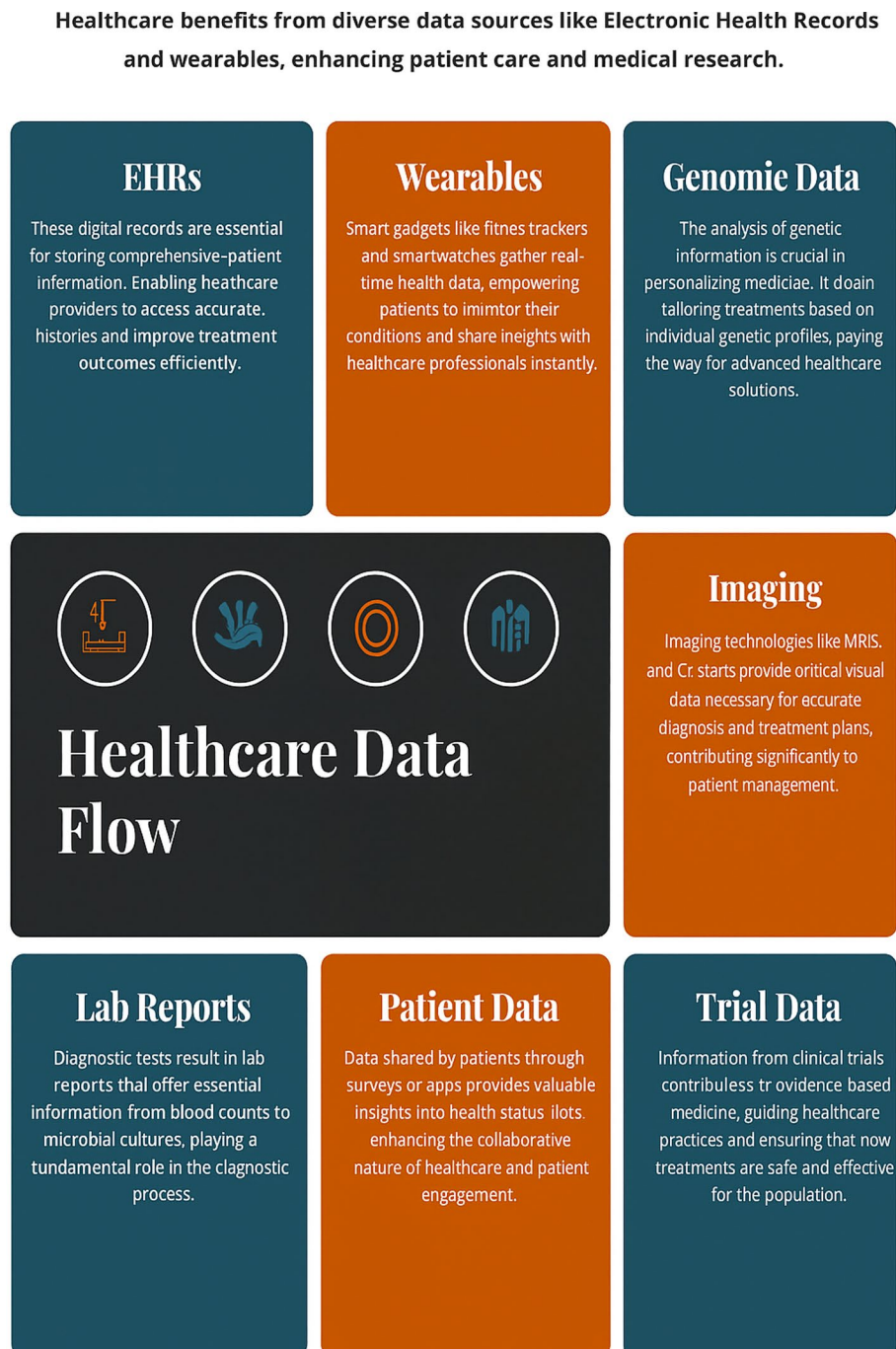
- (a) Operational Efficiency and Resource Optimization: In healthcare organizations, big data analytics play an important role to enhance functioning process in healthcare sector that can improve the various analytical facilities such as effective resource distribution, financial forecasting, man-power allocation, supply system etc. [62]. This operational efficiency enhances the cost estimation and optimizes the resources. Through data analytics, specialist can take comparatively more precise decisions, patient better treatment and over all enhance the healthcare system effectiveness [63]. Though, drawbacks of this system are also there such as security of history of patient's medical records and robust healthcare system framework. If we want to go with this data driven healthcare system and comprehended the healthcare ecosystem



**Fig. 3** Healthcare Predictive Analytics (created by the author)

to patients support, we have to handle these drawbacks and apply big data analytics effectively.

- (b) The increasing volume of data in healthcare: There are no of factors affecting for rising of clinical data in healthcare sector, some of them we are discussing here in detail. Certainly data formation, gathering and storing such a big deal in healthcare sector that has never been seen before. This scenario of data flood includes the digitalization of healthcare sector, regular monitoring with the help of gadgets, technology enhancement etc. [64]. Therefore data deluge of healthcare system has a great influence on patient care and hospitality, medicinal research and efficiency of the system [65]. The causes of data flooding in healthcare are depicted in Fig. 4.
- (i) Electronic Health Records (EHRs): Traditional clinical record system i.e. paper record system, in today scenario paper record has been replaced by Electronic Health Records (EHRs) that providing a digital warehouse for patient clinical record. These clinical data warehouse gives a comprehensive antiquity of patient-doctor communication, disease diagnosis and analysis, medicine recommendations, treatment history, psychoanalysis, healings, and additional significant clinical data records [66]. Improving patient outcomes, care coordination, and data accessibility is the aim of the move to electronic health records. Hence the application of EHRs has been predominantly accrued with exponential growth in healthcare clinical data [67].
- (ii) Medical Imaging: Developments in medical imaging technology, such as digital X-rays, CT scans, and MRIs, the quantity of diagnostic pictures has expanded dramatically. These high- resolution images are crucial for accurate diagnosis and treatment planning. They do, however, also significantly increase the total number of healthcare databases, requiring efficient techniques for storage and analysis.
- (iii) Wearables and Remote Monitoring: With the growing use of wearables and the Internet of Things (IoT), it is now feasible to continually monitor patient's health outside of traditional healthcare settings. Through the monitoring of vital signs, physical activity, and other health- related factors, these devices provide real-time data streams. The incorporation of data from remote monitoring into healthcare systems adds another layer to the already growing volume of health data [68] 66.
- (iv) Genomic and Molecular Data: An explosion of genetic and molecular data has been caused by the quick advancements in the domains of genomics and molecular medicine. Thanks to the ability to sequence whole genomes and analyze molecular profiles, personalized medicine and targeted pharmaceuticals now have additional possibilities. It is very typical task to examine, scrutinize, accumulate and understand these enormous and intricate datasets.
- (v) Healthcare Interoperability: It is the capability to share enormous amount of patient's data which is generated from disparate healthcare systems in a smooth way. Therefore interoperability provides more comprehensive and assimilated observation of particular patient's medical records among various healthcare systems that enhance the capability of data exchange facility [69].
- (vi) Patient-Generated Data: In today scenario patient generated data becomes highly convoluted to their personal care so that patient generated data is prominently upsurge. This includes information on lifestyle choices, symptoms, and self-reported outcomes. A comprehensive picture of a patient's health is provided by



**Fig. 4** The healthcare data flooding (created by the author)

combining this subjective data with clinical data, although doing so increases the volume of data overall [70].

- (vii) **Regulatory Requirements and Reporting:** Requirements for data collection, reporting, and compliance are imposed by regulatory organizations and healthcare facilities. The objectives of these laws are to guarantee patient safety, high-quality care, and compliance with medical guidelines. They do, however, add to the increasing amount of data that has to be handled and examined.

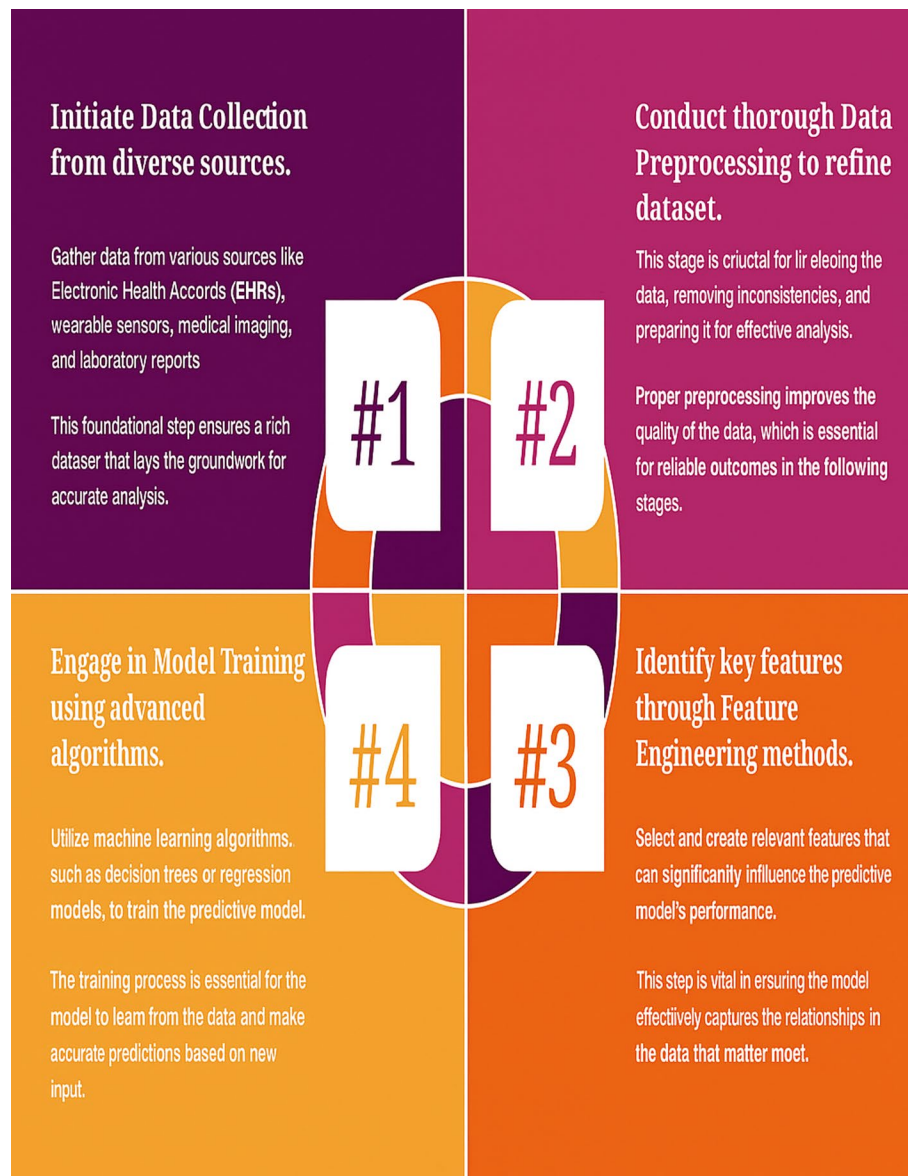


**Research and clinical trials:** Large volumes of data about treatment outcomes, intervention efficacy, and patient responses are produced by research studies, clinical trials, and observational studies. This study data contributes to the general body of knowledge about healthcare and is frequently preserved in computerized databases. The growing amount of medical data brings with it both advantages and disadvantages. Although data-driven insights and innovations have the potential to improve patient care, in order to extract useful and relevant information from this abundance of data, strong data management, security, and analytics skills are also required. Realizing the full potential of healthcare data becomes dependent on the healthcare industry's ability to effectively leverage new technologies, such big data analytics, as it navigates this data-rich world [71].

In the healthcare industry, predictive analytics refers to the application of sophisticated data analysis methods and predictive models to mine healthcare data for insightful information and forecast future results. Predictive analytics uses statistical modeling, historical data, and machine learning algorithms to help healthcare professionals make informed decisions, spot patterns and trends, and enhance patient outcomes. The following are some significant uses of predictive analytics in the medical field:

**Disease prediction and early Intervention:** Predictive analytics may be used to identify people who have a high risk of developing specific diseases, such as diabetes, cardiovascular disorders, or specific malignancies. By analyzing patient data, including demographics, medical history, genetic information, and lifestyle variables, predictive models may provide risk scores and probabilities [72]. It provides platform and make easy according to patient treatment plans, quick engagement and avoidance action. Figure 5 demonstrates the various applications of predictive analysis in the healthcare domain.

- (a) Hospital Readmission Forecast: Readmission in hospital may be very tidy and expensive, hence predictive analytics may recognize the influences that affect readmissions of patients, for instance treatment endurance, comorbidity, financial issues, with the help of assessing patient's medical record. The post discharge care, different care coordination's, remote healthcare etc. may support the healthcare specialists to decrease readmission proportions.
- (b) Patient Flow and Resource Management: One of another important role of predictive analytics in healthcare organizations is hospital operations to improve the estimation of patient's need, availability of space i.e. bed and different medical resources. The main characteristics of smart healthcare system is to ability to real time support of healthcare organizations on the basis of analysis of historical data, handle patient flow, medical staff availability and resource distribution, etc. These features improve the delay time, resource allocation and deliver better patient's requirement and hospitality.
- (c) Planning and Allocation of Resources in Public Health Emergencies: Healthcare predictive analytics play a crucial role in health related emergencies for instance in disease outbursts, natural calamities, etc. By analyzing real-time data, such as disease distribution patterns, population demographics, and healthcare capacity, predictive models can predict resource requirements, identify high-risk locations, and aid in decision-making for effective emergency response and resource allocation.
- (d) Clinical Decision Support: Accurate Predictive analytics may provide healthcare professionals with decision support tools by analyzing patient data, medical literature,



## Healthcare Predictive Analysis

**Fig. 5** Prediction analysis (created by the author)

and treatment outcomes. Based on available data, machine learning algorithms can build a predictive system that may predict the disease diagnosis, therapeutic activity, support medical specialist in decision making, enhance diagnosis precision, etc. Since with the help of predictive analytics we can empower the hands-on, data driven decision making healthcare environment. Healthcare professionals used this data driven decision making environment and may prescribe effectively. To implement predictive analytics effectively in healthcare sector, some hindrance must be resolved

such as data interruption, data confidentiality, data interoperability, assimilation, etc. [73].

#### 4 Issues and challenges in healthcare systems

Healthcare system is a system that provides the smart, better hospitality, remote healthcare and may handle critical intense situations for patient and pacify the healthcare demands over society. However, a lot of complications and lacking influences the proficiency, efficacy and approachability of such kind of systems [74]. Here we discuss some challenges that face by the healthcare systems in brief:

- (a) Access to Healthcare: Healthcare approachability is one of the most essential challenges in healthcare organizations that should make sure one and all have approach to high grade medical facilities. There is a lot of population those who belong to rural and remote areas, small town population faces the challenges to reach the high rated medical services. Many populations have some other issues also such as lack of money, reside in rural area, insufficient health insurance, etc. hence accessibility of high grade medical services universally is the necessity for entire population.
- (b) Growing Healthcare Costs: One of the most significant problems facing people is raising the cost of healthcare globally. The rising of population, diseases, enhancement of medical tools and technology are some of the reason behind the raising of cost of healthcare units. Therefore without losing the quality of medical healthcare hospitality, delivering the medical services is a immense challenge and issue for healthcare sector.
- (c) Quality and Patient Safety: Patient safety is a high priority concern for every healthcare unit without degrades the quality of patient treatment in healthcare system. However medical inaccuracies and variances in the features of healthcare taken in various healthcare systems are very harmful to any healthcare systems. Therefore proper observation and sustainable enhancement are necessary to resolve the current complicacies of medical inaccuracy, quality and enhancing the patient safety [75].
- (d) Health Information Systems: Effective patient care delivery depends on robust health information systems that facilitate the collection, sharing, and organization of patient data. Challenges include putting in place interoperable electronic health record (EHR) systems, safeguarding patient confidentiality and privacy, and making it simple for different healthcare providers and systems to exchange medical information. For healthcare companies, standardizing and integrating health information systems continue to be major challenges [76].
- (e) Aging Population and Chronic Diseases: As the world's population ages, chronic illnesses including cancer, diabetes, and cardiovascular disease are becoming more common. Global healthcare systems have significant problems in managing the intricate healthcare requirements of aging populations, offering suitable long-term care, and putting preventative measures in place to manage chronic illnesses.
- (f) Health Disparities and Inequalities: Numerous healthcare systems continue to exhibit health disparities and inequalities, whereby some communities are disadvantaged in terms of their health outcomes and have restricted access to medical care. These discrepancies are caused by socioeconomic variables, cultural barriers, racial and ethnic inequities, and inefficiencies in the healthcare system. Targeted treatments and policy changes are necessary to address health disparities and promote health equity.

- (g) **Integration of Technology:** For healthcare systems, the quick development of technology—such as wearables, telemedicine, and artificial intelligence—brings both benefits and difficulties. In order to effectively integrate technology into healthcare delivery, concerns like workforce training, legal frameworks, data privacy and security, and equal access to technology-enabled healthcare services must be addressed [77].
- (h) **Workforce Shortages and Burnout:** Physicians, nurses, and other allied health workers are among the healthcare professions that are frequently in short supply in healthcare systems. Burnout among healthcare workers is a result of long work hours, high stress conditions, and mounting expectations on healthcare professionals. For healthcare systems, addressing labor shortages, enhancing work-life balance, and advancing the wellbeing of medical workers are essential problems.
- (i) **Healthcare Policy and Governance:** Within intricate frameworks of governance and policy, healthcare systems function. For legislators and leaders in the healthcare industry, creating and executing efficient healthcare policies, maintaining regulatory compliance, and striking a balance between conflicting interests and goals are difficult tasks [78]. Healthcare governance frameworks must be open, responsible, and flexible enough to adapt to changing population demands.
- (j) **Global Health Challenges:** Global health issues include pandemics, infectious disease outbreaks, and natural disasters also face healthcare systems. Coordinated actions, efficient emergency planning, and robust healthcare infrastructures that can withstand demand spikes and deliver timely and appropriate care are all necessary for responding to severe disasters. Stakeholder cooperation, creative methods of delivering healthcare, legislative changes, infrastructural investments in the healthcare sector, and a patient-centered care philosophy are all necessary to meet these problems. Healthcare systems must be continuously observed, assessed, and improved upon in order to guarantee their resilience and capacity to satisfy the changing healthcare requirements of the populace.
- (k) **Data Security and Privacy:** The patient information is of inherently sensitive nature and therefore carries a high level of confidentiality attached to it, and as such is the primary target for all manner of cyberattacks. The application of cloud storage for such information is riddled with inherent dangers due to the high likelihood of unauthorized access to such vital information [79]. Coupled with this, the need to comply with regulatory requirements such as HIPAA and GDPR adds another level of complexity that makes it increasingly difficult to ensure strict security measures. Despite the fact that measures such as encryption and anonymization are absolutely essential for protecting such sensitive information, such measures therefore limit access to the information, and even slow the rate at which breakthroughs in machine learning technology can be realized [80].
- (l) **Computational Cost:** It requires a humongous amount of computational power to process the enormity of healthcare information that is constantly being generated, and this translates to outrageously high storage and real-time processing costs. This cost can prove particularly burdensome for small healthcare facilities, which may not be in a position to support the high costs that are incurred in handling such large amounts of information. Although distributed computing is a promising line of approach that can potentially mitigate some of such burdens involved in handling

such information, it simultaneously imposes additional complexities and challenges involved in its implementation that must be addressed effectively[81].

- (m) **Bias and Interoperability:** Artificial intelligence models have the potential to inherit biases from the historical data upon which they are trained, and this can lead to biased outcomes in healthcare across different demographic segments. Coupled with this challenge, the presence of varied data formats on varied systems makes integration of such information highly complicated. Consequently, there is an immediate and compelling need for the institution of standardization, as well as the development of effective strategies with a view to preventing bias. This is necessary in order to ensure that the insights obtained through artificial intelligence applications are sound and equitable, and that they promote fairness in healthcare outcomes.
- (n) **Ethical Challenges: Bias, Fairness, and Privacy-Preserving ML:** Healthcare ML models face ethical dilemmas beyond technical constraints. A key issue is bias in datasets, where underrepresentation of minority populations leads to unfair outcomes and exacerbates health disparities. Fairness in predictions requires transparent, interpretable models and fairness-aware algorithms that can actively detect and mitigate bias. Equally important is privacy protection. Traditional centralized training often risks exposing patient data, whereas federated learning enables collaborative training across hospitals without sharing raw data. Combined with differential privacy techniques, these approaches ensure compliance with global regulations (HIPAA, GDPR) while safeguarding patient trust. However, balancing privacy, fairness, and model performance remains an open challenge that future research must address.
- (o) **Data Quality and Heterogeneity:** Clinical information is often incomplete, noisy, or inconsistent due to variations in medical devices, human errors in manual recording, or absent values. This impacts the performance and reliability of ML models learned from such information [12].
- (p) **Interoperability Issues:** Hospitals' and clinics' Electronic Health Records (EHRs) are also in non-standard formats, and it is time-consuming and challenging to integrate them [13].
- (q) **Algorithmic Bias and Fairness:** ML models trained from biased data sets will continue to exacerbate existing health disparities and make erroneous predictions for minority or underrepresented groups [14].
- (r) **Real-Time Deployment Constraints:** The majority of ML algorithms are latency-prone and computationally intensive, thus limiting their application to time-critical clinical settings such as acute care [15].
- (s) **Interpretability and Trust:** Doctors are hesitant to adopt "black-box" models like deep learning systems that are uninterpretable or untransparent, a source of serious concern in high-stakes medical decision-making [15].
- (t) **Regulatory and Ethical Issues:** There is a lack of proper standardization of data governance, patient consent procedures, and data transfer between borders, creating legal uncertainties in the use of AI in healthcare [16].

Furthermore, each of the technologies in big data is strong and weak in a particular aspect. Table 6 indicates which of the technologies is weak in particular areas of healthcare.

As seen in the Table 6, technologies like TensorFlow offer excellent real-time capabilities but are weak in interpretability and integration. Conversely, Scikit-learn offer

**Table 6** Technology-specific limitations in healthcare applications

| Technology    | Data quality handling | Real-time analytics | Interpretability | Integration with EHRs | References |
|---------------|-----------------------|---------------------|------------------|-----------------------|------------|
| Hadoop        | Moderate              | Poor                | Low              | Poor                  | [12]       |
| Apache spark  | High                  | Good                | Moderate         | Moderate              | [13]       |
| TensorFlow    | Low                   | Excellent           | Very Low         | Limited               | [14]       |
| Scikit-learn  | Moderate              | Poor                | High             | Poor                  | [15]       |
| MLlib (Spark) | High                  | Good                | Moderate         | Moderate              | [13]       |
| AutoML tools  | Poor (generalized)    | Moderate            | Low              | Very Low              | [16]       |

transparency but lacks scalability and real-time analytics. These differences underscore the need for careful technology selection based on use-case, whether it's chronic disease monitoring, ICU decision support, or national health policy planning.

## 5 Predictive analytics using machine learning

To evaluate large amounts of data, identify trends, and provide accurate projections, predictive analytics heavily depends on machine learning algorithms [82]. A few well-liked machine learning algorithms for predictive analytics in a variety of fields, including healthcare, are as follows:

- (a) Linear regression: Linear regression is a basic and widely used method in predictive analytics. It illustrates the relationship between a dependent variable and one or more independent variables by fitting a linear equation. The application of linear regression makes it simpler to predict continuous outcomes, such as a patient's blood pressure, based on their age, weight, and other relevant data.
- (b) Logistic regression: Logistic regression is used when the result variable is binary or categorical. It determines the probability that an event will occur using the variables that are provided. Logistic regression is often used in medicine to predict a patient's likelihood of contracting a certain disease or the efficacy of a treatment plan.
- (c) Decision Tree: This algorithm is used for classification and regression tasks and has hierarchical configuration that separates data into root node, child nodes and leaf nodes. Decision Tree algorithm uses the attributes to split the data into more comprehensible cluster and may prune branches to avoid overfitting [83]. It is easy to visualize, interpretable and may handle missing values. In healthcare sector, Decision Tree algorithm are widely used for risk measurement, healing selection, prediction of missing values, illness diagnosis, etc.
- (d) Random Forest: Random Forest algorithm is a collaborative learning algorithm to enhance the precision and vigor of predictions. It uses decision tree as a base algorithm and then conglomerates various decision trees to increase performance and with the help of bootstrapping random forest diminishes the overfitting. It may handle huge and complex data sets and provide precise estimation. This model is widely used in healthcare sector to recognize intense critical situation of the patient, sickness identification.
- (e) Support Vector Machine (SVM): It is a supervised learning method which is used for both classification and regression jobs and SVM works well for both linear and non-linear separable data. SVM works for data preparation which is plotted in high dimensional space, hyperplane that separates data into classes, margin and support vectors. There are number of applications of SVM in healthcare sector such as



bioinformatics, medical image classification, disease classification, risk evaluation, etc.

- (f) **Neural Networks:** It is a machine learning model which contains layers of interconnected nodes i.e. neurons which process and spread information. Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) are types of neural network and have grown acceptability in healthcare industry since their capability to recognize multifaceted patterns and extract features from raw data. Sequential data such as time series, speech, memory, feedback loops, may be studied with Recurrent Neural Networks. On the other hand grid like data such as images X-Rays and MRIs, 2D signals, spatial hierarchies, convolution layers may be understood with the help of Convolution Neural Networks.
- (g) **Naive Bayes:** This machine learning algorithm is commonly used for classification tasks like spam recognition, image classification, text classification, sentiment analysis, etc. especially when working with enormous dataset. It is widely used in healthcare organizations particularly for email straining, illness classification, sentiment analysis, etc. and can handle categorical & numerical data. Naïve Bayes machine learning algorithm is based on Bayes theorem and assumes features are independent of each other than calculate the probability of each feature given the class label then multiply the probabilities of all features together and pick the class of highest probability.
- (h) **Gradient boosting:** Gradient boosting is an ensemble learning technique that builds a powerful prediction model by combining weak learners. It builds a series of models one after the other, each trying to address the shortcomings of the one before it. Two gradient boosting algorithms that are widely utilized in healthcare applications, such as illness prediction, readmission prediction, and customized therapy suggestion, are XGBoost and LightGBM [84]. These algorithms are renowned for their high predicted accuracy.

Predictive analytics uses a few of these machine learning algorithms [85]. The particular challenge, the properties of the dataset and the intended results all play a role in choosing the best algorithm. To find the best strategy for a particular predictive analytics activity, it's usual practice to test out several algorithms and evaluate how well they work [71] (Table 7).

## 6 Methodology

Utilizing a range of strategies and technology, smart healthcare systems improve patient outcomes, facilitate proactive and individualized treatment, and improve healthcare delivery. The following are some current methods and tools that are often employed in intelligent healthcare systems:

- (a) **Internet of Things (IoT):** The networking of devices and sensors enabled by Internet of Things technology makes real-time data collection and sharing feasible [95]. IoT devices may monitor patient vitals, handle medication adherence, and provide data to medical professionals for remote monitoring in intelligent healthcare systems. This real-time data facilitates early health problem detection, continuous patient condition monitoring, preventive treatments etc.[96].
- (b) **Wearable Devices:** Wearable electronics, such as fitness trackers, smartwatches, and biosensors, is being used by smart healthcare systems more and more. These devices

**Table 7** Analysis of various algorithms and technologies in big data analysis

| Algorithm               | Type                     | Advantages                                                                   | Disadvantages                                                                     | Big data suitability                                                              | Refs.    |
|-------------------------|--------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|
| Linear regression       | Guided learning approach | Straightforward, quick, and easy to interpret                                | Restricted to modeling solely linear patterns                                     | Extremely scalable for handling vast datasets                                     | [86, 87] |
| Decision trees          | Guided learning approach | Easily understood and capable of modeling non-linear trends                  | Vulnerable to issues of overfitting                                               | Moderately scalable in large data scenarios                                       | [88, 89] |
| Random forests          | Guided learning approach | Highly precise and resistant to overfitting                                  | Demands significant computational resources                                       | Effective for big data, though resource-demanding                                 | [88, 90] |
| Support vector machines | Guided learning approach | Provides exceptional accuracy and minimizes overfitting risks                | Incurs substantial computational expense                                          | Scalability is somewhat limited                                                   | [36, 91] |
| Neural networks         | Guided learning approach | Offers superior precision and high adaptability                              | Requires extensive datasets and high computing power                              | Exceptionally scalable, particularly with cloud or GPU support                    | [92]     |
| Naive Bayes             | Guided learning approach | Simple, rapid, and performs admirably with high-dimensional inputs           | Assumes that features are independent, which can be unrealistic                   | Extremely scalable and efficient with large-scale datasets                        | [93, 94] |
| Gradient boosting       | Guided learning approach | Delivers superior predictive accuracy, effective for structured/tabular data | Training is slow, risk of overfitting if parameters not tuned carefully           | Well-suited for big data but requires parallelization and optimized frameworks    | [90]     |
| Logistic regression     | Guided learning approach | Simple, interpretable, efficient for binary/multiclass classification        | Limited to linear decision boundaries, less effective for complex non-linear data | Highly scalable for large datasets, often used as baseline in healthcare ML tasks | [86]     |

monitor vital signs, activity, sleep patterns, and other health-related data. They provide helpful advice on how to manage long-term conditions, promote healthy lifestyles, and administer prophylactic care. Furthermore, wearable technology enables medical practitioners to keep an eye on patients' health outside of clinical settings and monitor them remotely [97].

- (c) **Electronic Health Records (EHR):** By digitizing patient medical records, EHR systems facilitate the efficient storage, retrieval, and sharing of patient data between healthcare facilities. Electronic health records provide a comprehensive image of a patient's test results, allergies, and medication history (EHRs). When EHRs are integrated with smart healthcare systems, patient data is readily available, care coordination is improved, and clinical decision-making is facilitated.
- (d) **Telemedicine and Telehealth:** Telemedicine and telehealth technology provide remote monitoring, virtual consultations, and telehealth platforms. These technologies make it easier for people to get healthcare, particularly those who live far away or have limited mobility. Telemedicine makes healthcare more accessible, decreases the need for in-person visits, and boosts patient satisfaction.
- (e) **Artificial Intelligence (AI) and Machine Learning:** Smart healthcare systems use AI and machine learning algorithms in many different ways. For the purposes of illness diagnosis, risk assessment, treatment planning, and customized medicine, they evaluate enormous datasets, spot trends, and forecast future events. Chatbots and virtual assistants driven by AI may also offer self-management assistance, triage services, and patient education.
- (f) **Data Analytics and Predictive Modeling:** The extraction of insights from healthcare data is made possible by data analytics approaches such as predictive modeling.

Predictive models use both past and current data to determine which patients are at high risk, forecast the course of a disease, and allocate resources as efficiently as possible [10]. Healthcare providers can make better judgments, run their businesses more smoothly, and treat patients better when they use data analytics.

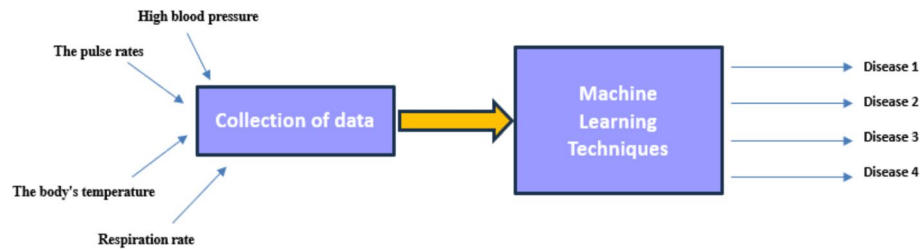
- (g) Remote Monitoring and Sensor Technologies: Remote monitoring technologies provide ongoing patient health monitoring outside of traditional healthcare settings. This includes platforms for remote patient monitoring, wearable technologies, and sensors put in homes [98]. By giving them access to real-time data on vital signs, medication adherence, and activity levels, these technologies enable healthcare professionals to remotely monitor patient conditions and respond proactively [99].
- (h) Blockchain Technology: Blockchain technology offers secure and decentralized storage and sharing of healthcare data. It enhances data security, interoperability, and privacy in smart healthcare systems. Blockchain can facilitate secure exchange of patient data, ensure data integrity, and enable patient-controlled access to their health records [100].
- (i) Robotic Process Automation (RPA): RPA automates routine administrative tasks, streamlining healthcare workflows and reducing manual errors. RPA can handle tasks like appointment scheduling, claims processing, and data entry, freeing up healthcare professionals' time for more patient-focused activities.
- (j) Big Data Analytics: Big data analytics is used by smart healthcare systems to manage massive amounts of healthcare data, such as those from real-time sensor data, medical imaging, genomics, and electronic health records. The discovery of population health patterns, research discoveries, and precision medical techniques are made possible by big data analytics [101].

Healthcare is a necessary yet costly sector. This is growing due to a variety of factors, including population growth and ageing, sickness prevalence and incidence, increased demand for health-care services, and increases in health-care and pharmaceutical costs. This upward trend in healthcare costs posed difficult challenges to healthcare systems in terms of efficiency and productivity, and was one of the driving factors behind a shift in this sector, which was previously primarily focused on hospital care, to place a greater emphasis on preventative measures [102]. Many problems can be diagnosed by engaging in various human activities such as walking, jogging, standing, or completing any work. For instance, our body naturally adjusts our heartbeat to suit our activities and the environment [103]. Consequently, our heart rates rise in times of stress, anxiety, or agitation and fall in times of comfort, quietness, or sleep. Heart rate is therefore a useful indicator of general health [104]. An unusually high or low heart rate may be a sign of cardiac or other health issues. Physicians may be able to identify health problems by feeling the pulse throughout the body [105]. An overview of the working of healthcare system is illustrated in Fig. 6.

Healthcare system will take data from the human's day to day activity and form data set for analysis. The collected data may be structured or unstructured, in next step this huge amount of data will analyze using machine learning techniques.

Data collection belongs to the following factors: -

- The body's temperature
- The pulse rates



**Fig. 6** Working of healthcare system (created by the author)

- Respiration rate
- High blood pressure

Now the collected big data first separate using topic modeling to form different clusters and then pass to prediction system to match with different symptoms. After analyzing data with the symptoms of disease system will generate probable disease [106].

A gadget that resembles a smart watch and fits in your hand gathers data for the forecast [107]. This gadget has sensors that gather data for the many parameters the prediction system needs. Data for the prediction will collect by following two ways:

- (i) IoT based wearable device.
- (ii) Additional data sources, including genetic databases, medical imaging, and electronic health records (EHRs).

### 6.1 Comparative analysis of available research in AI and IoT-based healthcare

The table presented in this report presents a detailed and comparative review of different research studies that have been centred on a wide range of applications of artificial intelligence (AI), the Internet of Things (IoT), and machine learning (ML) in specific in the healthcare sector. The table is effective in classifying these different studies based on major criteria that include focused areas of research, the specific technologies that have been utilized, and the major findings that have been obtained from the outcomes of each study.

#### (a) Early Research on AI and IoT in Healthcare

Zhu et al. [102] performed a study of human activity recognition using radar-based artificial intelligence. Bonazzi et al. (2017) performed a study of Wireless Body Area Networks (WBANs) with special emphasis on sensor-based monitoring systems in healthcare environments.

The findings obtained from these studies emphasized the role of wearable technology and non-contact monitoring systems, and in overcoming challenges related to sensor optimization as well as real-world applications of the same technologies.

#### (b) IoT and ML-Based Healthcare Solutions

Majumder et al. [108] focused their research on the application of IoT and machine learning technologies specifically with the aim of enhancing elderly care and support. Fortino et al. [109] performed a study of cloud-integrated IoT systems and API-based healthcare services that can enhance patient care.

Yang et al. [110] performed a study of e-health services using Google Cloud Messaging (GCM) but presented the limitations that exist in obtaining real-time analytics, especially in medical cases of emergency.

(c) Developments in IoT-Driven Healthcare Frameworks

Habibzadeh et al. [63] and Farahani et al. [111] evaluated the capability of IoT devices that have been integrated with artificial intelligence and machine learning for the purpose of real-time decision-making processes.

Rodrigues et al. [99] and Mshali et al. [112] discussed the significant role that the Internet of Health Things (IoHT) can play in contemporary healthcare delivery.

Kumar and Devi Gandhi [113] presented a study of a three-tier IoT architecture that has been developed specifically for the early detection of heart disease, using both AI technologies and Apache Big Data.

(d) Recent Studies (2020–2025) on AI, IoT, and Big Data Analytics

Remote Patient Monitoring & Telemedicine: performed an investigation of the central role that 5G, IoT, and artificial intelligence can play in.

- Edge Computing for Real-Time Analytics: In their 2022 published paper, Smith et al. referred to the critical role played by edge computing in healthcare, citing its crucial role in reducing latency and thus improving decision-making in high-risk scenarios like critical care.
- Cyber security in IoT Healthcare Systems: In a 2023 proposal by Johnson et al., there was a new data-centric approach, the aim of which was to improve the security of IoT-habituating cloud environments in the healthcare industry, responding to the increasing need for effective protection.
- Big Data Analytics in Healthcare: Lee et al. [114] revealed their work in 2021, which described the vast advantages that accrue with the convergence of heterogeneous data sources. Their work referred to how such convergence can result in enhanced predictive analysis and personalized healthcare options.

## 6.2 Key takeaways

- (a) The healthcare sector is evolving at a fast pace with the advances in AI and IoT technologies, which are empowering capabilities like real-time monitoring, predictive analysis, and new cloud-based solutions that greatly improve patient care and operational efficiency.
- (b) Notwithstanding these advances, however, there are still a series of challenges that must be addressed, including on-going security issues, the requirement for effective data management strategies, and the requirement for seamless integration of real-time applications to ensure an integrated healthcare experience.
- (c) The shift from conventional healthcare practices to those that are increasingly AI- and IoT-based is obviously clear. In order to realize the full potential of these technologies, however, there must be more improvement in infrastructure, interoperability, and the creation of all-encompassing regulatory regimes.

This Comparative Analysis of existing research is briefly described below with the help of analysis table i.e. Table 8.

**Table 8** Comparative Analysis of Available Research in AI and IoT-Based Healthcare

| No. | Topic/reference title and publication                                       | Technology used                   | Key insights/purpose                                                                                                         | Remarks/observations                                                                                          | References |
|-----|-----------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------|
| 1   | Indoor human activity recognition [102]                                     | HARAR radar, ML                   | Employs radar technology to recognize human activities, eliminating the need for wearable sensors                            | Demonstrates reliable activity recognition; however, it has not undergone comprehensive real-world evaluation | [115]      |
| 2   | Wireless body area network (WBAN) (Bonazzi et al. 2017)                     | WBAN, sensor networks             | Provides an overview of WBAN applications in healthcare while addressing challenges in sensor network implementation         | Underscores the growing importance of WBAN in healthcare, though sensor designs need further optimization     | [116]      |
| 3   | IoT and machine learning techniques [108]                                   | IoT, ML                           | Centers on merging IoT with machine learning to enhance elderly healthcare, highlighting obstacles and progress in the field | Addresses elderly care use cases, yet falls short of real-time performance validation                         | [108]      |
| 4   | Cloud-based IoT in healthcare                                               | Cloud IoT, API-based systems      | Explores cloud-based API solutions to unify multiple healthcare services                                                     | Stresses the importance of interoperability, though it does not include a practical case study                | [117]      |
| 5   | IoT, web of things (WoT) (G. Fortino et al. [109])                          | IoT, WoT, RESTful web services    | Introduces a cloud-oriented framework designed for remote monitoring of healthcare                                           | Covers remote healthcare monitoring, but offers limited details on implementation                             | [109]      |
| 6   | E-health services for elderly [110]                                         | Google cloud messaging (GCM), ICT | Leverages Google Cloud Messaging to deliver push notifications for elderly healthcare monitoring                             | Efficient in delivering alerts, yet lacks capabilities for real-time data analysis                            | [110]      |
| 7   | Future Internet and IoT [76]                                                | IoT, large-scale systems          | Examines the growth of IoT sensors and the associated challenges in expansive healthcare applications                        | Addresses scalability challenges, but does not provide specific healthcare application examples               | [118]      |
| 8   | A survey of healthcare IoT devices: a clinical perspective [63]             | IoT devices, AI                   | Evaluated IoT applications in healthcare with a focus on communication efficiency and computing limitations                  | Introduces decision-support terminology, though it falls short in offering insights into real-time sensing    | [119]      |
| 9   | Towards collaborative IoT for health (Farahani et al. [111])                | IoT, AI, ML                       | Investigated frameworks combining IoT and AI to facilitate healthcare data exchange                                          | Presents a well-organized model; however, clarity regarding real-time analytics is absent                     | [111]      |
| 10  | Enabling technologies for the Internet of Health Things (IoHT) [99]         | IoT, IoHT                         | Examines issues of data security, communication protocols, and cloud integration in healthcare systems                       | Examines security issues, but does not detail healthcare-specific implementations                             | [120]      |
| 11  | A novel three-tier IoT architecture for early heart disease detection [113] | ML, apache big data, AI           | Describes an AI-driven solution employing machine learning for early detection of heart disease                              | Primarily centered on heart disease detection, with limited application to broader healthcare scenarios       | [113]      |
| 12  | A survey on health monitoring systems for smart homes (Mshali et al. [112]) | IoT, AI, ML                       | Evaluates healthcare monitoring systems that utilize AI to support elderly care and manage chronic diseases                  | Brings attention to issues regarding the reliability of raw data processing                                   | [112]      |



**Table 8** (continued)

| No. | Topic/reference title and publication                                                                                             | Technology used                           | Key insights/purpose                                                                                                               | Remarks/observations                                                                                                 | References |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------|
| 13  | 5G and IoT for intelligent healthcare: AI and machine learning approaches—a review (2024, Springer) [58]                          | 5G, IoT, AI, ML                           | Examines how combining 5G, IoT, AI, and ML can improve remote patient monitoring and telemedicine services                         | Highlights both advantages and obstacles associated with integrating advanced technologies in intelligent healthcare | [58]       |
| 14  | Cloud computing, artificial intelligence, and machine learning in healthcare: the future of patient care (Popli 2024, IJFMR) [59] | Cloud computing, AI, ML                   | Investigates the role of cloud computing and AI in revolutionizing patient care, diagnostic processes, and medical research        | Stresses enhancements in data security, reduced costs, and increased accessibility in healthcare                     | [59]       |
| 15  | Edge computing for real-time healthcare analytics: opportunities and challenges                                                   | Edge computing, AI, ML                    | Assesses edge computing's contribution to real-time healthcare analytics, reducing latency and facilitating prompt decision-making | Identifies challenges including limited resources and security vulnerabilities at the network edge                   | [121]      |
| 16  | Data-centric framework for digital health: securing IoT devices in a cloud environment                                            | IoT, cloud, cybersecurity, data analytics | Introduces a data-centric framework aimed at integrating and safeguarding IoT-derived data in digital health environments          | Concentrates on counteracting cyber threats and securing dependable data transmission for robust remote monitoring   | [122]      |
| 17  | Big data and chronic disease management                                                                                           | Big data analytics, IoT, ML               | Explores how big data analytics can enhance predictive analysis                                                                    | Illustrates the potential of data-driven strategies to enhance clinical outcomes                                     | [123]      |

## 7 Framework of proposed prediction system

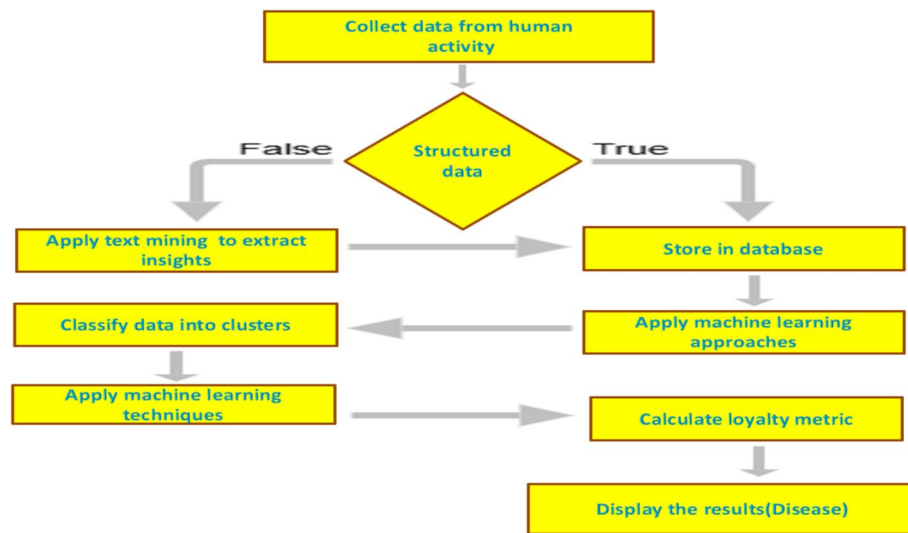
In this review paper there is no use of Clinical Trial even Data set. The framework of proposed healthcare system that incorporates machine learning algorithms involves following key components and the block diagram of the proposed prediction system framework is illustrated in Fig. 7.

### (A) Data acquisition and preprocessing:

- Determine whether electronic health records (EHRs), medical imaging, wearable technology, and genetic databases are pertinent sources of information.
- Having a robust data acquisition pipeline to collect and aggregate data from various sources.
- To guarantee consistency and eliminate any noise or outliers, clean, preprocess, and normalize the data.

### (B) Feature selection and extraction:

- Identify the relevant features or variables that will be used as inputs to the machine learning algorithms.
- Perform feature engineering to convert unstructured data into features that are useful and gather pertinent information.



**Fig. 7** Framework of Proposed Prediction System (created by the author)

- Select the most informative features using techniques like statistical analysis or feature importance ranking.

(C) Algorithm selection:

- Identify the appropriate machine learning algorithms based on the healthcare problem (e.g., classification, regression, clustering).
- Train and fine-tune the selected algorithms using the preprocessed data.
- Analyze several models performances with suitable metrics and cross-validation methods.

## 8 Discussion

The results of this research illustrate ways in which big data analytics and machine learning (ML) technologies can enable more productive, accurate, and unbiased healthcare decision-making at scale. Mention must be made, however, of assumptions made, observations drawn, areas for improvement, and limitations faced along the way so that the results can be interpreted and utilized in the proper sense.

### 8.1 Assumptions

A conceptual model of assumptions informed the study design and interpretation of results.

In the beginning, datasets used in this study were assumed to be representative of the target population of the patients but demographic groups were not reasonably represented well [124].

Second, semantic alignment of heterogeneous data—i.e., electronically formatted health records (EHRs) to unstructured clinical narratives and wearable device data—was conducted with the expectation that semantic alignment and harmonization of meta-data would decrease variation between sources.

Third, the ML models were also supposed to generalize in novel healthcare settings, provided careful procedures for identifying bias, feature choice, and cross-validation were used.

Finally, it was presumed that performance gains observed in the health care sector could, to a large extent, be replicated in other sectors too with similarly humongous amounts of data.

## 8.2 Lessons learned

The efforts at research did bear some fruit.

Second, bias removal is itself an ongoing endeavor and not a process which has to be appended on merit alone in the entire ML pipeline—from preprocessing to model fitting to post-hoc fairness analysis [125]. Model explainability was as important to clinician uptake; XAI methods, such as SHAP values, not only facilitated increased confidence but facilitated clinical verification of predictor variables. But another key finding was that the difficulty of fusion of the data increases disproportionately with the size of the set of heterogeneous data sources and requires scalable schema-matching and semantic-enrichment infrastructure.

Lastly, multiple exposures to the health workers during the modeling process produced stronger and more clinically relevant prediction models, and as evidence of the strength of domain expert feedback loops.

## 8.3 Recommended changes

We can deduce from these results some of the implications of future research and use:

Improve representational populations for data collection towards more equity and representativeness. Leverage real-time data streams to support real-time, dynamic decision-making.

Describe hybrid modeling methodologies integrating data-driven ML with rule-based domain reasoning to guarantee explainability for high-stakes decision-making systems. This is to be achieved through continuous bias auditing using visual analytics dashboards for continuous model validation.

## 8.4 Weaknesses

In spite of the encouraging results, there were some research limitations.

The biggest problem was the quality and amount of the data, especially for the under-represented clinical and demographic subgroups, which can limit the models' generalizability [126].

Apart from this, computational power was not able to accommodate training of very large-scale deep models and therefore there was dependency on less computationally demanding models. Interoperability, privacy, and governance of cross-institutional data sharing also placed a limit on how large the data aggregation could be. Finally, retrospective data sets were employed to perform this work and therefore prospective, real-world testing in live healthcare environments must be done prior to large-scale release.

## 9 Conclusion

Big data analytics (BDA) and machine learning (ML) are reshaping healthcare by enabling early diagnosis, personalized treatment, and better utilization of limited resources. This survey consolidates current progress, highlighting how these technologies strengthen decision-making, support precision medicine, and address persistent challenges such as heterogeneous data, limited predictive power, and inefficient resource

allocation. By synthesizing prior studies, this research provides a structured framework that can guide further innovation and practical deployment of BDA and ML in healthcare.

### 9.1 The findings of this research can be beneficial to different stakeholders directly

- **Researchers:** This study offers a consolidated knowledge base and comparative analysis of algorithms, helping identify gaps in explainability, bias minimization, and integration of multimodal data for future studies.
- **Practitioners (Healthcare Professionals):** Clinicians can adopt ML-driven decision support systems to improve diagnostic accuracy, design personalized treatment regimens, and manage chronic diseases more effectively.
- **Policymakers and Public Health Authorities:** By using predictive models, health agencies can track population-level disease trends, allocate resources efficiently, and strengthen national responses to crises such as pandemics.
- **Technology Developers & Industry:** The highlighted gaps and opportunities provide directions for building scalable, transparent, and secure AI-powered health platforms.
- **Patients:** Ultimately, patients benefit through faster diagnoses, proactive care, and improved quality of life.

### 9.2 Highlighting contributions and practical implications

This paper makes several contributions. First, it synthesizes diverse BDA and ML approaches in healthcare and evaluates their comparative performance. Second, it identifies common challenges such as poor data quality, weak interpretability, and ethical concerns, while suggesting frameworks to mitigate them. Third, it highlights actionable implications—researchers can refine models, practitioners can deploy clinically reliable tools, and policymakers can design fair, data-driven health strategies. Collectively, these contributions bridge the gap between academic development and real-world implementation.

### 9.3 Potential application of findings to other fields and industries

Although healthcare is the main focus, the approaches reviewed—such as bias auditing, explainable AI, and predictive modeling—are transferable across domains that face large-scale, heterogeneous data. For example, public health and epidemiology can benefit from outbreak forecasting; finance and insurance can apply predictive models to risk management and fraud detection; smart cities and urban planning can improve resource allocation and emergency response; agriculture can leverage predictive analytics for crop yield forecasting; and education can apply fairness-aware models in student performance prediction. Across these domains, the emphasis on fairness, scalability, and transparency provides a universal framework for data-driven decision-making.

## 10 Future scope

Gazing out into the expanse of possibilities on the horizon, the potential that lies for the use and application of big data analytics in conjunction with machine learning in the healthcare space is enormous, offering a range of promising avenues and opportunities for future research studies:

### 10.1 Advanced predictive models

Future research and studies can analyze and investigate next-generation algorithms that include advanced methodologies—like deep learning, reinforcement learning, and hybrid models—to increase predictive accuracy and allow for the development of more sophisticated and personalized treatment plans for patients. These advanced models can leverage increasingly heterogeneous datasets, from genomics to patient-generated data, to predict disease progression more accurately.

### 10.2 Real-time and edge analytics

With the rapid evolution of technologies like 5G and the advent of edge computing, there is huge scope and potential for the development of systems that can perform and analyze data in real-time. This phenomenal capability is critical for a huge range of applications, from remote patient monitoring to early intervention in critical care situations, and rapid response mechanisms in case of health emergencies. Research studies in this particular area can be focused on reducing latency and enhancing the scalability and effectiveness of real-time analytics.

### 10.3 Improved data integration and interoperability

As the volume and variety of healthcare data keep growing exponentially, the development of robust frameworks for efficiently integrating heterogeneous data sources will become absolutely necessary. Future work should be focused on developing standardized protocols and interoperable systems that enable seamless and efficient data sharing across platforms, enhancing the comprehensiveness and overall utility of healthcare analytics for a diversity of stakeholders.

### 10.4 Robust security and privacy solutions

With the inherently sensitive nature of healthcare data, further research on advanced security measures becomes a necessity and an imperative. This includes examining blockchain technology as a means of secure data exchange, as well as the development of advanced encryption and anonymization techniques that maintain patient privacy without sacrificing the accessibility of data for analytical purposes.

### 10.5 Ethical and bias mitigation strategies

As artificial intelligence models become more common and widespread in various clinical settings, it is of the highest importance that these inherent biases found in these systems are confronted and addressed systematically. It is imperative that these technologies result in fair outputs for all patients, regardless of background or situation. In the future, research efforts should be aimed at a great extent at the development of algorithms that are not only transparent but also fair, while simultaneously working on developing ethical guidelines that can effectively function to promote trust and accountability in AI-based healthcare systems.

With regard to efficiency in operations and optimization of resources, predictive analytics has the potential to play a pivotal and revolutionary role in optimizing resource allocation within the intricate structure of healthcare systems. Future research efforts have the potential to delve deeper into the studies of how big data insights can be seamlessly integrated into hospital management systems, potentially resulting in

improvement in financial forecasting, streamlined workflow processes, and an overall improvement in overall operational efficiency across the board.

When taken together, these various paths and avenues underscore the promise of a future where the intersection of leading-edge analytics, the processing of real-time data, and the implementation of robust security measures will not only enhance patient outputs but also usher in a revolution in healthcare delivery. This revolution is aimed at providing a more proactive, efficient, and personalized service to all those in need of care.

#### **10.6 Directions for future research.**

Future research in medical care and machine learning and big data analysis should focus on the following topical directions:

(a) Advanced Predictive Modeling Techniques

- Create and test next-generation algorithms like deep learning, reinforcement learning, and hybrid AI systems.
- Highlight the application of integration of multiple sources of data like EHRs, genomics, sensors, and patient-reported outcomes to guide highly individualized treatment plans.

(b) Real-Time and Edge Analytics

- Describe the use of 5G and edge computing for real-time health data processing at the point of care.
- Such applications as remote monitoring, critical care decision support, and emergency response systems.

(c) Interoperability and Standardized Integration

- Create frameworks and APIs that facilitate seamless communication and data sharing among heterogeneous healthcare systems.
- Promote the use of standardized ontologies and protocols to enable harmonized patient records.

(d) Security and Privacy Preservation

- Explore technologies like federated learning, differential privacy, and blockchain for privacy-enhancing and secure data sharing.
- Maintain patient confidentiality while retaining the analytical value of healthcare data sets.

(e) Bias Minimization and Ethical AI

- Highlight the development of explainable and transparent AI models for clinical use.
- Place high priority on eliminating algorithmic bias to give unbiased results to different demographic and socioeconomic classes.
- Establish moral standards for the use and management of AI in healthcare.

(f) Operational Efficiency and Resource Optimization



- Explore how predictive analytics can be used to improve hospital operations, including staff scheduling, resource allocation, and cost management.
- Implement large data software in hospital information systems to gain real-time insights into administrative processes.

(g) Cross-Disciplinary Collaboration

- Foster collaboration among technologists, data scientists, policymakers, and clinicians to make sure innovations are ethically suitable, clinically viable, and consistent with healthcare goals.

Author contributions Conceptualization, analysis of available research in AI & IoT, Analysis of various algorithms and technologies in Big Data Analytics. Investigating, writing original draft, writing review & editing Healthcare System.

(h) AI–IoT–Blockchain Integration: The convergence of artificial intelligence with IoT-enabled healthcare devices and blockchain technology can create secure, real-time, and decentralized healthcare ecosystems. IoT wearables can continuously monitor patient vitals, AI can analyze these data streams for predictive insights, and blockchain can ensure immutable, transparent, and tamper-proof medical records. Future research should focus on scalable frameworks that integrate these three technologies for personalized and trustworthy healthcare delivery.

(i) Federated Learning for Privacy Preservation: Federated learning offers a novel approach to privacy-preserving ML by allowing multiple hospitals or clinics to collaboratively train models without exchanging raw patient data. This ensures compliance with regulations such as HIPAA and GDPR while maintaining the benefits of large-scale predictive modeling. Future studies should investigate efficient federated architectures and communication protocols to balance model accuracy, security, and scalability.

(j) Explainable Machine Learning (XAI) for Transparency: To foster clinician trust and adoption of AI-driven tools, explainability must be prioritized in healthcare ML models. Explainable AI (XAI) methods can provide interpretable outputs that clarify how predictions are generated, particularly in sensitive tasks like disease diagnosis and treatment planning. Research should emphasize hybrid models that balance predictive power with transparency, ensuring that AI remains an assistive and accountable tool in clinical decision-making.

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**Data availability**

All data generated or analysed during this study are included in this published article.

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